

The Partner Predicate

Incidence, Forcing, and the Inverse Problem for BBP Relation Spaces

Driven by Dean A. Kulik

August, 2026

Every numerical claim below was produced by code executed in the session that generated it. Residuals are quoted from live output. Section 12 is the falsification ledger: what was killed, by which measurement, and what replaced it. Nothing in this paper is carried forward from an earlier draft without re-measurement.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Abstract

A BBP frame (b, N) generates shell sums S_r and a kernel K of integer vectors annihilating them. This paper reports one predicate governing that kernel, verified in both directions on the same object, and a structural split it produces between lineages.

The predicate: given a distinction inside a structure, ask whether it has a partner it can cancel against. With no partner it is forced; with a partner it is free. Applied to prime ideals in the orbit decomposition of $x^N - b$, this single question derives both classical seed kernels with no integer-relation search and excludes 1893 of 2233 frames by factorisation alone with zero false exclusions. Forcing is established universally within the stated domain. Occupancy — which freed orbits actually carry weight — is measured for the octal subdivision lineage and does not generalise across lineages; the hexagonal counterexample is given.

The invariant that makes the predicate bidirectional is not the prime content but the incidence structure: the map sending each prime to the set of orbits carrying it, with multiplicity. Incidence one means no partner and forces vanishing; incidence two or more means a partner and leaves the coordinate free. Prime labels alone are insufficient, and the paper exhibits a frame where reasoning from labels gives the wrong answer.

The inverse problem is solved to the extent it can be. Given only a coefficient vector, the orbit factors are separated from the free completion by an intrinsic test — orbit factors are equimodular, with all roots on the circle of radius $|b|^{1/N}$, and the free completion is not. That radius recovers $|b|^{1/N}$ directly. What the vector determines is $U(b)$, the unshared part of $b-1$; b itself is not recoverable, because a shared prime leaves no forced factor and is invisible to the reconstruction.

The gating result is new, and it separates into two steps that the paper had been treating as one. A prime dividing the period ramifies in $Q(\zeta_N)$; the forcing lemma's hypothesis fails there, and the coordinate is freed. That step is derived and universal. Whether a freed coordinate is then occupied is a second question, measured here for the octal subdivision lineage: an unshared channel opens if and only if its prime divides the period. A control frame $(16, 56)$ carries a prime dividing the period with no corresponding channel and opens nothing, and the roles of 3 and 5 reverse on command between $N = 24$ and $N = 40$, confirmed as a forward prediction on a previously uncomputed frame.

Freed is not occupied. Three frames are freed and unoccupied, including $(64, 42)$, where an unshared prime divides both $b - 1$ and the period, the frame is a genuine subdivision, and the channel stays shut. The mechanism of freeing is settled; the mechanism of occupancy is not.

The lineage split is the deeper result. The octal lineage has no shared channel and its relation dimension gates: 1,1,5,1,5,1 across $q = 1, 2, 3, 4, 5, 7$. The hexagonal lineage has one shared channel and its dimension counts: 1,2,3,4 across $q = 1, 2, 3, 4$. The same predicate, applied where a shared channel exists and where it does not, produces a switch in one lineage and an accumulator in the other.

Also reported: a third seed at negative base, a grading theorem determining when class-pure detection is complete, a ceiling theorem naming the terminal coefficient field of any frame before computation, and the correction of $\dim K$ at $(16, 24)$ from 5 to 3, which retires the phase-correction quantity the corpus has carried.

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1. The Predicate

Stated once, in the general form, before any application:

**Given a distinction inside a structure, ask whether it has
a partner it can cancel against.**

No partner -> forced. Partner -> free.

That is the whole rule. Everything in this paper is that sentence at a different location. The paper is organised so that each section names the location and reports the measurement, rather than renaming the predicate.

The predicate is bidirectional and both directions are used:

- Forward. Partner structure decides which orbits are forced, hence which coefficients must vanish, hence the shape of the relation.
- Backward. Which orbits are forced decides the partner structure, hence the prime content, hence the frame.

Two distinct statements shelter under that one sentence and the paper separates them. The forcing predicate — incidence one implies forced — is derived in section 4 from valuations, under the hypothesis that p does not divide N . The occupancy predicate — a freed channel carries weight — is measured in section 9. Section 9.5 shows the two are joined at one end and open at the other: the hypothesis of the forcing lemma fails exactly when p divides N , so freeing is derived, while occupancy is not.

Section 11 gives the inverse procedure and states exactly where it succeeds and where it cannot.

2. Setup, Prior Art, and Continuity

2.1 Definitions

Fix an integer base b with $|b| > 1$ and a period $N \geq 2$. The shell sums are

$$S_r = \sum_{k \geq 0} 1 / ((Nk + r) b^k), \quad r = 1 \dots N-1$$

The $r = 0$ shell diverges and is excluded. The kernel K is the lattice of integer vectors c with $\sum c_r S_r = 0$. Write $C(y) = \sum c_r y^r$, so $C(0) = 0$ by construction.

Let $\theta = b^{1/N}$, $\omega = \zeta_N$, and $\alpha_j = \theta \cdot \omega^j$ for $j = 0 \dots N-1$, the roots of $x^N - b$. Put $\beta_j = 1 - 1/\alpha_j$ and $L_j = \log(\beta_j)$ on the principal branch.

Factor $x^N - b = \prod g_i$ over the integers. For monic g of degree d , $\prod (1 - \alpha) = g(1)$ and $\prod \alpha = (-1)^d g(0)$, so the orbit norm is $g_i(1)/g_i(0)$.

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2.2 Prior art

The closed form on which the reduction depends is not new. Borwein, Borwein and Galway, *Canad. J. Math.* 56 (2004) 897–925, give at their equation (64)

$$L(1, b, n, j) = -(1/n) b^{(j/n)} \sum_r \zeta_n^{(-rj)} \ln(1 - \zeta_n^r b^{(-1/n)})$$

which after the substitution $\alpha = b^{(1/n)} \zeta_n^{(-r)}$ becomes $S_r = -(1/n) \sum_j \alpha_j^r \log(1 - 1/\alpha_j)$. Verified here to 120 digits on (16,8), (64,6), (4096,12), (4096,24). Their valuation-vector method (their section 3.2) is the origin of the forcing machinery used below, and they themselves note its limitation: knowledge of the p-adic data determines the imaginary part of a logarithm only modulo 2π , so valuations return potential dependencies.

Their Research Problem 10 asks whether these techniques extend to fixed (b, n) , and observes that this would require a good understanding of $\mathbb{Q}[\zeta_n, b^{(1/n)}]$. Section 8 supplies part of that understanding. No claim is made here to answer Problem 10.

2.3 Continuity with the prior corpus

Kernel, Frame, Winding, Residue (July 2026) states the thesis of this paper in its section 9.7: the null lattice is a property of the shell constants alone, exists whether or not any relation-detection algorithm is run, and PSLQ selects a representative without creating the class. Structure is prior; the interface samples it. That paper also derives both seeds from angle admissibility — $\cos 45$ and $\cos 60$ — rather than by sweep, and derives the coefficient vector $\{4, -2, -1, -1\}$ from the integral representation.

Four of its stated open problems are answered by measurements in this paper:

Open problem	Statement	Resolved by
Omega 2	rank of the null lattice for general n	dim R measured on 19 frames (section 9)
Omega 3	do n=16,24 carry relations not reducible to unrolled windings	surplus table (section 7.3)
Omega 4	crossing bases at 4096; earlier null over a small slice	dim R = 7 at (4096,24) (section 9)
Omega 5	which constants appear in which frame	orbit-prime incidence (section 9.1)

Its Omega 5 records the octal frame yielding $\log 2$, $\log 3$, $\log 5$ and the hexagonal yielding $\log 3$ and $\log 7$. That is reproduced here from orbit norms: unshared primes $\{3, 5\}$ at octal, $\{7\}$ at hexagonal with 3 shared.

3. Reduction: the Kernel is a Relation Lattice

Using the closed form, $\sum_r c_r S_r = 0$ becomes $\sum_j C(\alpha_j) L_j = 0$. The coefficients $C(\alpha_j)$ are algebraic and the L_j are logarithms of algebraic numbers, so Baker's theorem applies: a vanishing linear form in

logarithms of algebraic numbers with algebraic coefficients forces the coefficient vector into R tensor $Qbar$, where R is the Q -space of relations among the L_j . Hence

$$K = \{ c : (C(\alpha_j))_j \text{ in } R \text{ tensor } Qbar \}$$

and R is a multiplicative relation lattice among the β_j inside the S -unit group of $Q(\zeta_N, \theta)$ — the field BBG named. R is cut out by two independent families: valuation conditions, one per prime ideal, and argument conditions from the unit and root-of-unity part. Valuations alone overcount; both together give R exactly.

3.1 Two consequences

Lemma 3.1 (sum-zero). Every c in R satisfies $\sum_j c_j = 0$.

Proof. Let p divide b . Then $v_p(\alpha_j) = v_p(b)/N$ is independent of j since the ω^j are units, and $v_p(\alpha_j - 1) = 0$, so $v_p(\beta_j) = -v_p(\alpha_j)$ is the same for every j . The valuation condition at p reads $-\sum_j c_j = 0$. Verified on every relation vector computed in this work.

Theorem 3.2. $\dim K$ over $Qbar$ equals $\dim R$.

Proof. Over $Qbar$ the map $c \rightarrow (C(\alpha_j))_j$ is injective with image the span of the columns (α_j^r) , $r = 1..N-1$. Its one-dimensional annihilator is spanned by the all-ones vector, since the power sums $\sum_j \alpha_j^r$ vanish for $1 \leq r \leq N-1$. So the image is exactly $\{v : \sum v_j = 0\}$, which contains R tensor $Qbar$ by Lemma 3.1, and the intersection is all of it.

Together with Baker, the ladder of readers terminates at $\dim R$ and cannot exceed it. $\dim R$ is the frame-intrinsic quantity; $\dim K$ over any particular field is a readout.

4. Forcing: the Seed Kernels Derived

This section obtains both classical kernels with no PSLQ, no coefficient bound, and no precision threshold, so none of the instrument failures catalogued in section 12 can reach them.

Forcing Lemma. Suppose $v_p(b-1) = 1$ and p does not divide N . Then p divides exactly one $g_i(1)$, to first power; exactly one prime ideal above p has nonzero valuation on the β_j ; it has residue degree 1 and divides exactly one β_j . The valuation condition at that ideal forces the corresponding coordinate to zero, and since R is Galois-stable, the coordinate vanishes on the whole orbit. Therefore every kernel polynomial C is divisible by g_i .

The hypothesis p does not divide N is required and was initially missing. Section 12 records its discovery.

4.1 Octal, (16,8)

$\theta = 16^{1/8} = \sqrt[4]{2}$, which lies in $Q(\zeta_8)$, so the field is $Q(\zeta_8)$ of degree 4, unit rank 1 with fundamental unit $1 + \sqrt[4]{2}$, and 2 is totally ramified. Writing $s = \sqrt[4]{2}$ and $z = \zeta_8$:

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```

beta_0 = u^-1 s^-1   beta_4 = u s^-1
beta_1 = z s^-1   beta_7 = z^-1 s^-1
beta_2, beta_6 : conjugate primes over 3 <- unshared
beta_3, beta_5 : conjugate primes over 5 <- unshared

```

So coordinates 2,3,5,6 vanish, and on {0,1,4,7} the conditions give $c_0 = c_4$, $c_1 = c_7$, $\text{SUM } c = 0$. Hence $R = \text{span}\{(1, -1, 0, 0, 1, 0, 0, -1)\}$ and $\dim R = 1$. C is therefore divisible by $y(y^2+2)(y^2+2y+2)$, leaving one free quadratic. Solving gives a single free parameter and

$$C = [8, -8, -4, -8, -2, -2, 1]$$

the Bailey–Borwein–Plouffe formula, with series residual 4.9×10^{-80} . $\dim K = 1$ is forced, not measured.

4.2 Hexagonal, (64,6)

$\theta = 64^{1/6} = 2$, so the field is $Q(\zeta_6) = Q(\sqrt{-3})$, degree 2, unit rank zero. Units are torsion only, so the argument conditions do real work here.

j	beta_j	(v2,v3,v7,v7bar)	arg
0	1/2	(-1,0,0,0)	0
1	$(3+i\sqrt{3})/4$	(-1,1,0,0)	$\pi/6$
2	$(5+i\sqrt{3})/4$	(-1,0,1,0)	irrational
3	3/2	(-1,2,0,0)	0
4	$(5-i\sqrt{3})/4$	(-1,0,0,1)	irrational
5	$(3-i\sqrt{3})/4$	(-1,1,0,0)	$-\pi/6$

Valuation conditions give $\text{SUM } c = 0$, $c_1 + c_5 + 2c_3 = 0$, $c_2 + c_4 = 0$ — dimension 3. Argument conditions give $c_1 = c_5$ and $c_2 = c_4$ — dimension 1, generator $(-1, 1, 0, -1, 0, 1)$, residual 8.7×10^{-62} . The two-dimensional gap between them is exactly the layer BBG say valuations cannot see. Proceeding as before yields one free parameter and

$$C = [16, -24, -8, -6, 1]$$

with series residual 4.1×10^{-80} .

4.3 Consequence

The statement that the principal relation annihilates exactly the unshared-prime orbits, previously carried as a verified observation, becomes a one-line corollary: an unshared prime has nothing to cancel against, so its coordinate must vanish. The count of such orbits is a rank deficiency, not an auxiliary statistic.

5. Prime Content and the Exclusion Sieve

Since $\text{PRODUCT}_i g_i(1) = 1 - b$, the odd-prime content of a frame is the factorisation of $b - 1$, distributed by N and never created by it. The published winding table, presented as evidence that winding manufactures primes, is the factorisation of $b^k - 1$ — Cunningham and Zsigmondy, and BBG's own Theorem 3 (Bang) is the exclusion tool built from it.

5.1 $v_p(b-1) = 1$ implies unshared; the converse fails

Confirmed on all nine rows of the published table: 16 gives $15 = 3 \cdot 5$ and unshared $\{3, 5\}$; 64 gives $63 = 3^2 \cdot 7$ with 3 shared and 7 unshared; 4096 gives $4095 = 3^2 \cdot 5 \cdot 7 \cdot 13$ with $\{5, 7, 13\}$ unshared; 1048576 gives $3 \cdot 5^2 \cdot 11 \cdot 31 \cdot 41$ with 5 shared, which is the row where the table records 5 going shared.

The converse is false. Searching n in $[3, 12]$ and b in $[2, 400]$ produced 20 counterexamples — primes sitting in exactly one orbit with multiplicity greater than one. Example: $(64, 4)$, where $b-1 = 63 = 3^2 \cdot 7$ and 3 occupies a single orbit at exponent 2. Also $(100, 4)$, $(289, 4)$, $(361, 4)$, $(243, 5)$, $(100, 6)$, $(64, 8)$. Only the implication survives, and every downstream use takes the safe direction.

5.2 The sieve

Corollary (necessary for $\dim K > 0$). With the forcing lemma, C is divisible by y and by g_i for every orbit carrying a prime with $v_p(b-1) = 1$ and p not dividing N , so

$$1 + \text{SUM over such orbits of } \deg g_i \leq N - 1$$

Factorisation only — no PSLQ, no precision, no coefficient bound. Over b in $[2, 320]$ and N in $[3, 9]$:

quantity	value
frames examined	2233
rigorously excluded	1893 (84.8%)
false exclusions against 16 frames of known nonzero $\dim K$	0
survivors	340, of which the reducible cases are the only ones that can carry

Before the ramification hypothesis was added the count was 1952. The 59 withdrawn exclusions are precisely those that leaned on a prime dividing N ; they form a mathematically identifiable class, not a scatter of errors. The sieve is necessary and not sufficient, and it is silent on $(16, 12)$.

6. The Grading Theorem

Let $d = [Q(\zeta_N, \theta) : Q(\zeta_N)]$. For $b = 2^m$ this is computable in closed form: write $m/N = p/g$ in lowest terms, so $\theta = 2^{(p/g)}$; then $d = g/2$ if g is even and 8 divides N , since $\sqrt{2}$ lies in $Q(\zeta_N)$, and $d = g$ otherwise.

Theorem 6.1. For any u in the orthogonal complement of R with rational entries,

$$\sum_j u_j C(\alpha_j) = \sum_r c_r \theta^r T_r(u), \quad T_r(u) = \sum_j u_j \omega^j(jr)$$

Every $T_r(u)$ lies in $Q(\zeta_N)$. Since $\{1, \theta, \dots, \theta^{(d-1)}\}$ is a $Q(\zeta_N)$ -basis and θ^d lies in $Q(\zeta_N)$, the condition splits into d independent conditions indexed by $r \bmod d$. Hence K decomposes as a direct sum over residues mod d .

Corollary. When d equals the subdivision index q , every kernel element is class-pure and class-pure detection is complete rather than a probe. When $d < q$ nothing forces purity and mixed relations may exist that a class-pure detector cannot see.

6.1 Winding raises the period

A parameter error propagated through the corpus and inverted several readings. Winding maps (b, n) to (b^m, n) — it raises the period as well as the base, which is exactly why $x^{(nm)} - b^m$ contains $x^n - b$ as a factor. Therefore $(4096, 24)$ is the octal seed wound $k = 3$ with $q = 1$. It is not a $q = 3$ subdivision and no character or carrier apparatus applies to it.

frame	n	q	d	d vs q	dim K	class-pure sum	gap
(16,16)	8	2	2	$d = q$	1	1	0
(16,24)	8	3	3	$d = q$	3	3	0
(16,32)	8	4	4	$d = q$	1	1	0
(64,24)	6	4	2	$d < q$	1	1	0
(4096,12)	6	1	1	$d = q$	1	0	1
(4096,24)	8	1	1	$d = q$	5	0	5

6.2 The phase-correction quantity does not exist

The corpus reports a correction of 2 at exactly $(16, 24)$ and $(4096, 24)$ and treats it as the central mystery. At $(16, 24)$, $\dim K$ is 3, not 5. Two independent methods agree: the full 23-shell iterated PSLQ returns exactly three relations, stable from $\max\text{coeff } 10^5$ to 10^7 at residual near 10^{-300} , and the class-by-class decomposition

returns $1+1+1$. All three are class-pure, as Theorem 6.1 requires since $d = q = 3$. The gap there is zero and cannot be otherwise.

At $(4096, 24)$ the gap is real but it is 5, not 2, and none of the five relations is class-pure because $d = 1$ and nothing forces purity. The carriers previously reported at that frame were read through the winding map rather than measured in the frame. The quantity is therefore not a phase correction; it is the failure of the grading, and it is provably zero whenever $d = q$.

7. Transport, Ancestry, and a Third Seed

7.1 Winding is transport, not generation

With $x^{24} - 4096 = (x^8 - 16)W$ and $W = y^{16} + 16y^8 + 256$, the product $W \cdot D_{\text{seed}}$ has residual 1.34×10^{-297} against the shells of $(4096, 24)$. So the inclusion holds. But $\dim K$ at that frame is 5 while the image of the seed is one-dimensional, so the inclusion is strict. The identity $D_{\text{frame}} = W \cdot D_{\text{seed}}$ is false; what is demonstrated is that the two operations act on distinct coordinates of the construction.

Lift heights are exactly deterministic — source height times cofactor height. Octal into $(256, 16)$: $8 \times 16 = 128$. Octal into $(4096, 24)$: $8 \times 256 = 2048$. Hexagonal into $(4096, 12)$: $24 \times 64 = 1536$. Height therefore tracks inheritance depth and nothing else; a conjecture that height measures how underconstrained a frame is died on measurement, since the shortest vector at $(4096, 24)$ has height 256, below both inherited lifts.

7.2 The ancestor family, closed

A frame (bo, m) feeds (b, N) when $bo^k = b$ and $m \mid q^k = N$, with lift $C(y) = W_k(y) D(y^q)$, subdivision first then winding. The real integer solutions of $bo^k = b$ are $bo = \text{plus or minus } b^{(1/k)}$, negative only for even k . That family is finite, enumerable, and complete within the class of frames $x^m - bo$ with integer bo .

The enumerator used to produce the corpus contained a guard rejecting roots below 2. The entire negative half of the family was structurally unreachable. It was not a mathematical gap; it was an authored one. The same error class recurred later in a root-finding routine that returned on first match, so negative roots were found only when the positive one failed.

7.3 The third seed

$4096 = (-64)^2$, and $x^{24} - 4096 = (x^{12} - 64)(x^{12} + 64)$. The right factor is the frame $(-64, 12)$, present in the factorisation throughout. Its kernel has dimension 2:

```
[ 0, 16, -16, -24, 0, 8, 4, 6, 6, 1, -1 ] height 24 residual 1.1e-300
[ 32, -32, -32, 0, -8, -16, -4, 0, -4, -2, 1 ] height 32 residual 3.6e-301
```

Every ancestor of $(-64, 12)$ is empty: $(-64, 6)$, $(-64, 4)$, $(-64, 3)$, $(-64, 2)$ via $k = 1$, and $(-4, 4)$, $(-4, 2)$ via $k = 3$. Both relations are native to it. It carries dimension 2 where both classical seeds carry 1, and its coefficient heights are smaller than the octal seed's. The sweep that produced the two-seed claim ran b in $[2, 320]$ and n in $[3, 9]$; $(-64, 12)$ is outside on both axes.

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Including it, the inherited span at (4096,24) is 4 rather than 2, and the residual native dimension is 1, not 3. The corpus figure of 3 arises from two cancelling errors: an ancestor span of 2 instead of 4, and a total of $\dim K = 5$ instead of $\dim R = 7$.

8. The Reader Ladder and the Ceiling

$\dim K$ over a given coefficient field is a readout. The holdout frame (16,12) reads zero over Q , yet $\dim R = 1$ and the relation is

$$8t S_1 - 16 S_2 - 8t S_4 + 4 S_5 + 2t S_7 - 4 S_8 - 2t S_{10} + S_{11} = 0, \quad t = 2^{1/3}$$

with residual 5.1×10^{-260} . The frame is full; the reader was rational. This retires the ledger line that treated the zero at (16,12) as evidence of absence.

frame	Q	$Q(\theta)$	$Q(\theta, \sqrt[3]{2})$	$\dim R$
(64,6)	1	1	—	1
(16,8)	1	1	—	1
(16,12)	0	1	—	1
(16,24)	3	3	5	5
(4096,24)	5	—	7	7

At (16,24) the relation is homogeneous of θ -degree 2: every term $c_r \theta^r$ reduces to a rational multiple of θ^2 , and the field-component of each coefficient is fixed by $r \bmod 3$. The rational reader misses one graded component, not a scattered set of algebraic numbers.

Ceiling Theorem. Every element of K over $\overline{Q}$ is recovered by inverse DFT, $c_r = \lambda w_{\hat{r}} / (N \theta^r)$ with w in Q^N . So $w_{\hat{r}}$ lies in $Q(\zeta_N)$, θ^r in $Q(\theta)$, and every relation coefficient lies in $Q(\zeta_N, b^{1/N})$ up to one global scalar — a field determined by (b, N) alone, before any shell is computed.

Consequently the reader ladder is bounded in advance and no distributed component can occur in a BBP frame. Observed coordinates confirm it: $2^{1/3}$ at (16,12), $2^{1/6}$ and $\sqrt[3]{2}$ at (16,24), $\sqrt[3]{2}$ and $\sqrt[3]{2}$ at (4096,24) — all inside the predicted ceiling, and $\sqrt[3]{2} = \zeta_{12} + \zeta_{12}^{-1}$ lies in $Q(\zeta_{24})$ for both $N = 24$ frames. The ceiling is a bound, not an address: it does not say which sub-field a given direction occupies, and that must be computed direction by direction.

9. The Gate

This section reports the new result. The forcing lemma says an unshared prime forces its orbit to vanish. The question left open is when such an orbit can instead carry weight.

9.1 Orbit incidence

The operative quantity is incidence: the map sending a prime p to the set of orbits whose $g_i(1)$ it divides, written m_p for its cardinality. Incidence, not $v_p(b-1)$, is what the forcing argument uses, and it is the invariant that makes the predicate bidirectional. A prime with $m_p = 1$ has no partner and forces its orbit to vanish; a prime with $m_p \geq 2$ can cancel against itself and leaves its coordinates free.

The distinction is not visible in the prime labels. Both lineages contain the prime 3; at octal it has $m = 1$ and forces, at hexagonal it has $m = 2$ and does not. Any procedure reasoning from labels rather than incidence will confuse the two, and section 11 exhibits a reconstruction that does exactly that and returns a wrong frame.

```
octal (16,8)  b-1 = 15 = 3 * 5
x^2 - 2    g(1) = -1  bins {0,4}  unit
x^2 - 2x + 2 g(1) = 1  bins {1,7}  unit
x^2 + 2    g(1) = 3   bins {2,6}  3-channel, m = 1
x^2 + 2x + 2 g(1) = 5   bins {3,5}  5-channel, m = 1

hex (64,6)  b-1 = 63 = 3^2 * 7
x - 2      g(1) = -1  bins {0}    unit
x + 2      g(1) = 3   bins {3}    3-channel
x^2 - 2x + 4 g(1) = 3   bins {1,5} 3-channel -> m = 2, SHARED
x^2 + 2x + 4 g(1) = 7   bins {2,4} 7-channel, m = 1
```

9.2 The two-factor gate

Measured across the octal lineage, where every channel is unshared:

frame	q	3 N	5 N	dim R	sym	anti	occupied bins mod 8
(16,16)	2	no	no	1	1	0	{0,1,4,7} unit only
(16,32)	4	no	no	1	1	0	{0,1,4,7} unit only
(16,56)	7	no	no	1	1	0	{0,1,4,7} unit only
(16,24)	3	YES	no	5	3	2	{0,1,2,4,6,7} + 3-channel
(16,40)	5	no	YES	5	3	2	{0,1,3,4,5,7} + 5-channel

The rule: an unshared channel opens if and only if its prime divides the period. The base decides which channels exist; the period decides which are open. Neither factor alone predicts anything.

(16,56) is the control and it is the decisive one. $56 = 8 \cdot 7$, seven is prime and divides the period — and nothing opens, because 7 does not divide $b - 1 = 15$ and there is no seven-channel in this frame. $\dim R = 1$, unit bins, identical to (16,16) and (16,32). This kills the tempting reading that $N = 8p$ with p prime is the mechanism: 24, 40 and 56 all have that form and only two of them do anything.

9.3 The reversal

The rule makes an exact, reversible prediction. At (16,24), 3 divides the period and 5 does not, so the 3-channel {2,6} opens and the 5-channel {3,5} is forbidden. At (16,40) the divisibilities exchange, so the roles must exchange.

Run on a previously uncomputed frame, the prediction held on the first attempt. Both frames return $\dim R = 5$ with the identical signature of 3 symmetric and 2 antisymmetric directions, differing only in which channel opened. The four dimensions contributed by an open channel are therefore a property of ramification, not of the particular prime.

The asymmetry that the corpus carried as an open problem — why the octal seed admits a 5-selective twisted shape and not a 3-selective one — is answered by this. It was never about 3 versus 5. It is about which prime divides the period, and it reverses on command.

9.4 Where the gate does not apply

Three frames appear to satisfy the gate and do not open: (16,12), (4,6), (4,12). All three have an unshared prime dividing the period and $\dim R = 1$. The missing condition is that the frame be a subdivision of a seed: 8 does not divide 12, and $b = 4$ supports no seed at all. The gate applies to subdivisions of a seed frame, not to arbitrary (b,N) .

9.5 Freed is not occupied

The gate and the forcing lemma are joined at one end. The forcing lemma of section 4 carries the hypothesis that p does not divide N , and section 12 records why: when p divides N the prime ramifies in $Q(\zeta_N)$, the conjugate prime ideals in the compositum are no longer distinct, and the valuation condition permits cancellation within the orbit rather than forcing each coordinate to zero. That is not an auxiliary technical condition. It is the mechanism.

So the first step of the gate is derived, not measured:

$p \mid N \rightarrow \text{ramification} \rightarrow \text{forcing hypothesis fails} \rightarrow \text{coordinate FREED}$

This is universal within the stated domain and follows from the same valuation argument that produces the forcing lemma. The measured content of section 9 is the second step, and it is a different question:

$\text{FREED} + ? \rightarrow \text{OCCUPIED}$

Freeing removes a prohibition. It does not supply weight. The distinction is visible in the data:

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frame	p	p N	freed	seed subdivision	occupied	dim R
(16,24)	3	yes	yes	yes	YES	5
(16,40)	5	yes	yes	yes	YES	5
(16,12)	3	yes	yes	no (8 does not divide 12)	no	1
(4,6)	3	yes	yes	no (no seed at b = 4)	no	1
(4,12)	3	yes	yes	no (no seed at b = 4)	no	1
(64,42)	7	yes	yes	yes	no	1

Three of the six freed channels are unoccupied because the frame is not a subdivision of any seed, which is the restriction already stated in section 9.4. The sixth, (64,42), is the one that matters: it is a genuine hexagonal subdivision at $q = 7$, the prime 7 is unshared with incidence one, 7 divides $b - 1 = 63$ and 7 divides the period, so the channel is freed by the same argument that frees the octal channels — and it stays shut.

The gap is therefore narrower and better located than it appeared. Freeing is a valuation fact and is settled. Occupancy requires something beyond freeing, that something is present in the octal lineage and absent in the hexagonal one, and the structural difference between those lineages is the presence of a shared channel, which section 10 measures independently.

10. Lineage Character

The deeper result is that the two lineages do not merely gate differently. They scale differently, and the difference is determined by whether a shared channel is present.

lineage	shared channels	q = 1	2	3	4	5	7	behaviour
octal (16,·)	0	1	1	5	1	5	1	gates
hexagonal (64,·)	1	1	2	3	4	—	1	counts

The octal dimension spikes: it is 1 unless an unshared prime divides q , in which case it is 5. The hexagonal dimension climbs monotonically with q through 1,2,3,4 and then drops to 1 at $q = 7$, where the shared prime's divisibility no longer supplies the increment.

The hexagonal occupancy is frozen in the same measurements. Bins {0,1,3,5} — the unit orbit plus the entire 3-channel — are occupied at every q tested, from 2 through 7, without exception. The 7-channel {2,4} never opens, even at (64,42) where 7 divides both $b - 1$ and the period and the orbit is unshared. The two-factor gate is an octal-lineage law with a clean counterexample one lineage over.

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Read through the predicate, the two lineages are one statement:

- An unshared channel has one orbit, so nothing can cancel against it, so it must be gated. Where gating decides, the address set moves with the period and the count is fixed.
- A shared channel has two or more orbits, so it can cancel against itself, so it needs no key and is open at every subdivision. Where nothing gates, the address set is fixed and only the count can move.

Octal has no shared channel and behaves as a switch. Hexagonal has one and behaves as an accumulator. The same predicate, applied at two locations with different partner structure, produces opposite readouts. The disagreement between the lineages is not a failure of the law; it is the law's content.

This locates the occupancy gap of section 9.5. The freed channel at (64,42) sits in a lineage whose shared 3-channel already occupies bins {1,3,5} at every subdivision tested. A plausible mechanism is that the cancellation budget in a lineage containing a shared channel differs from one without, leaving nothing for the freed channel to carry. That is not measured and is listed in section 13 as the principal mechanism gap.

11. The Inverse Problem

The forward direction takes a frame to a relation. This section asks the inverse: given only a coefficient vector, what can be recovered? The answer is partial, and the boundary is sharp.

11.1 A reconstruction that fails

The obvious procedure is to factor C, collect the primes appearing in $g(1)$ of each factor, and read the prime content of the frame off them. On the octal vector it works:

```
OCTAL c = [8, -8, -4, -8, -2, -2, 1]
y      g(1) = 1 no prime
y^2 - 4y + 2  g(1) = -1 no prime
y^2 + 2      g(1) = 3 prime 3
y^2 + 2y + 2  g(1) = 5 prime 5
product 3 * 5 = 15 -> b = 16 CORRECT
```

On the hexagonal vector the same procedure returns the wrong frame:

```
HEX c = [16, -24, -8, -6, 1]
y      g(1) = 1 no prime
y^2 - 8y + 4  g(1) = -3 prime 3
y^2 + 2y + 4  g(1) = 7 prime 7
product 3 * 7 = 21 -> b = 22 WRONG, true b = 64
```

Two separate errors are visible. The factor $y^2 - 8y + 4$ carries the prime 3 and is not an orbit of $x^6 - 64$ at all; it is the free completion. And the shared prime 3, which does occur in $b - 1 = 63$ with multiplicity 2, leaves no forced factor and is therefore invisible to any procedure reading forced structure.

The octal case succeeded only because its free completion happens to carry no prime. That is an accident of that seed, not a property of the construction.

11.2 The equimodularity lemma

Lemma 11.1. Let g be an irreducible factor of $x^N - b$ over the integers. Every root of g has modulus $|b|^{1/N}$. Consequently an orbit factor is equimodular, and a factor of C whose roots have distinct moduli is not an orbit.

Proof. The roots of $x^N - b$ are $b^{1/N} \omega^j$ with ω a root of unity, all of modulus $|b|^{1/N}$; any factor collects a subset of them.

This supplies the intrinsic discriminator the reconstruction needs. It requires no knowledge of b or N , only the coefficient vector, and it separates forced orbits from free completion in exactly the cases where prime labels do not:

```
OCTAL
y^2 + 2 roots +- i sqrt2 modulus 1.4142136 EQUIMODULAR -> orbit
y^2 + 2y + 2 roots -1 +- i modulus 1.4142136 EQUIMODULAR -> orbit
y^2 - 4y + 2 roots 2 +- sqrt2 0.5858, 3.4142 not -> free

HEX
y^2 + 2y + 4 roots -1 +- i sqrt3 modulus 2.0 EQUIMODULAR -> orbit
y^2 - 8y + 4 roots 4 +- 2 sqrt3 0.5359, 7.4641 not -> free
```

The common modulus of the orbit factors is $\rho = |b|^{1/N}$. This is a stronger recovery than the prime route offers: it gives the base directly from the geometry of the roots rather than by multiplying prime labels, and it excludes the free completion even when the free completion carries a prime.

11.3 The corrected procedure, and its limit

Given a coefficient vector c and the period N :

1. factor $C(y) = \sum c_r y^r$ over the integers
2. discard the factor y (the $r = 0$ exclusion)
3. keep factors whose roots are equimodular with common modulus $\rho \neq 1$
-> these are the forced orbits; $\rho = |b|^{1/N}$
4. discard the remainder -> free completion, even if it carries primes
5. product of primes in $g(1)$ over kept factors -> $U(b)$
6. $\rho^N \rightarrow |b|$

Clause 3 excludes the unit circle because a relation may contain cyclotomic factors, whose roots are equimodular at $\rho = 1$ and are not orbits of $x^N - b$. One of the two directions at $(-64, 12)$ exhibits this and returns $\rho = 1$ without the exclusion.

Applied to every coefficient vector produced in this work:

vector	N	orbit factors	rho	rho^N	b	U(b)	free carries
(16,8) octal	8	2	1.4142136	16	16	15	—
(64,6) hex	6	1	2.0	64	64	7	3
(-64,12) dir 2	12	3	1.4142136	64	64	65	—

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vector	N	orbit factors	rho	rho^N	b	U(b)	free carries
(-64,12) dir 1	12	3	1.0	1	64	65	— (unit circle)
(16,24) class 0	24	2	1.122462	16	16	15	—
(4096,24) lift	24	6	1.4142136	4096	4096	4095	—

What is recovered: ρ , hence $|b|$ given N ; and $U(b)$, the unshared part of $b - 1$. What is not recovered: $b - 1$ itself. At the hexagonal seed $U = 7$ while $b - 1 = 63$; the factor of 9 belongs to the shared prime and leaves no forced structure. The vector determines the unshared part of the frame's prime content and nothing more.

This is the honest form of the bidirectional claim. Forward, incidence decides which orbits are forced. Backward, the forced orbits — identified by equimodularity, not by prime labels — recover the incidence-one part of the prime content and the base radius. The shared part is structurally invisible in that direction, and the paper claims no more.

The free completion is worth naming rather than discarding silently. At octal it is $y^2 - 4y + 2$, at hexagonal $y^2 - 8y + 4$; in both cases it is the parameter the constraint system solves for once the forced vanishing is imposed. Reading it as an emergent object created by interaction is an error recorded in section 12.

12. Falsification Ledger

Everything killed, with attribution. Instrument failures are listed separately because the pattern is itself a finding.

12.1 Corpus claims killed by measurement

claim	status	evidence
$\dim K(16,24) = 5$	3	two independent methods, stable maxcoeff 1e5 to 1e7
phase correction = 2 at exactly two frames	does not exist	zero at (16,24) by Thm 6.1; 5 at (4096,24)
$D_{\text{frame}} = W \cdot D_{\text{seed}}$ identity	false	inclusion strict; quotient nonzero
three-layer map strictly invertible	false	inverse undefined on the quotient
exactly two seeds	false	(-64,12), $\dim K = 2$, all ancestors empty
(4096,24) is a $q = 3$ subdivision	false	winding raises the period; $q = 1$
(16,12) has no relation	false	exhibited over $\mathbb{Q}(2^{1/3})$, residual 5.1e-260
unshared iff $v_p(b-1) = 1$	converse false	20 counterexamples, e.g. (64,4)

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claim	status	evidence
$5 = 2$ inherited + 3 native at (4096,24)	two cancelling errors	span 4 not 2; dim R 7 not 5; native 1
carrier count formula	unproven	proposed as a repair, never derived or tested
class-o at (4096,72) not divisible by W	unmeasured	carried from corpus, not reproduced
vector-to-frame closes in general	witness only	hex vector returns $b = 22$, true $b = 64$
forced orbits = factors carrying primes	false	$y^2 - 8y + 4$ carries 3 and is not an orbit

12.2 Claims originated and killed in the same session

claim	how it died
both seeds are exactly the Kummer-degenerate frames	every perfect n-th power in the sweep collapses the field and carries nothing
height measures how underconstrained a frame is	shortest vector at (4096,24) has height 256, below both inherited lifts
$\dim K \text{ over } Q(\theta) = \dim R$	true only where the antisymmetric part is empty; false at (16,24)
$\dim K \text{ over } Q = \dim R+$	false at (4096,24): 5 against 3
native = 3 at (4096,24)	1, after negative-base ancestors
$\text{corank}(V) = \dim R^{\text{Gal}}$	held at 12 frames, failed at (-64,12)
7 is the hidden hexagonal prime inside the octal relation	7 appears at 12 of 32 evaluation points; one prime among twenty-two
the mirror transformations are symmetries of the zero set	$ \text{Aut}(P^1, Z) = 1$; all 210 ordered triples, identity only
$y^2 - 4y + 2$ is a fused factor created by interaction	it is the free parameter the constraint solve returns
$N = 8p$ with p prime is the gate	(16,56) opens nothing
the two-factor gate is a general law	fails on the hexagonal lineage at (64,42)
SHA's compression fiber is a distributed merge	no coordinate found in seven vocabularies; that is a fact about the vocabularies

12.3 Instrument failures

Every one of these produced a plausible number rather than an error.

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failure	effect	fix
real nullity computed where rational nullity required	over-reported dim K	grade by theta-powers; test rationality
global phase fixed at 1 in the interpolation	2 of 5 directions reported unrealisable	free the phase; antisymmetric needs $i \cdot C$
guard rejecting roots below 2	negative ancestors invisible; native inflated	$bo = \text{plus or minus } b^{(1/k)}$
root routine returning on first match	same error, one turn later	enumerate all integer roots
int() applied to rational nullspace vectors	silent truncation	clear denominators
counting how many basis vectors have a property	basis-dependent measure of a basis-free quantity	compute the invariant subspace directly

Pattern. Every forward claim in 12.1 and 12.2 — a quantity measured and reported — died. Every backward derivation — a constraint on what a relation must satisfy — survived, and two became derivations of the BBP formula rather than searches for it. Forward reasoning produces a witness and no information about what else exists; backward reasoning produces a condition on everything, so the count comes free.

Second pattern. The failures fall into distinguishable modes with distinct remedies: a broken probe needs a positive control; a wrong baseline needs a derived null; a narrow reader needs an independently computed ceiling; a wrong address needs a second route; a basis-dependent count needs the invariant computed directly. The taxonomy is not decoration — each mode is invisible to the controls that catch the others.

13. Open Problems

- THE OCCUPANCY MECHANISM. Freeing is derived: $p \mid N$ ramifies, the forcing hypothesis fails, the coordinate is released. What supplies weight to a freed coordinate is not derived. It is present in the octal lineage at (16,24) and (16,40) and absent at (64,42), where every stated condition is met. This is the principal mechanism gap.
- THE INVERSE-INFORMATION GAP. The vector determines $U(b)$ and the radius ρ . Whether shared-prime multiplicity is recoverable from any feature of C , or is genuinely absent from the relation, is open. These two gaps are of different kinds: the first is a mechanism question, the second an information question.
- Whether the hexagonal count law extends. Four data points fit $\dim R = \gcd(q,12)$; with five points that is a fit, not a law.
- Both channels ramified. Requires $8 \mid N$ and $15 \mid N$, so $N = 120$: predicted $\dim R = 9$, sym 5, anti 4, all eight bins occupied. 119 shells is past PSLQ reach here.
- The remaining native dimension at (4096,24). The four degree-4 irreducible factors of $x^{24} - 4096$ are not of the form $x^m - bo$; algebraic-base ancestors are the next representation rung.
- Seed classification. Not more examples: characterise the complete set of primitive generators. The negative half of the integer-base domain has never been swept and has already produced one seed of dimension 2.

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- The $p \mid N$ regime of the forcing lemma. Three known relation-bearing frames sit in it, all with $p = 3$ and orbit norm 3, and forcing fails in all three. Whether that reflects ramification or the scarcity of the regime is untested.
- Re-measurement. $(16,48)$, $(16,72)$, $(256,48)$, $(4096,36)$, $(4096,72)$ are all unmeasured with the corrected machinery. Every dimension in the prior corpus came from a detector that was wrong by 2 at $(16,24)$ and by 2 in the other direction at $(4096,24)$.

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Appendix A. Methods

Shell evaluation. Direct summation to $\text{ceil}(3.5 \text{ dps} / \log_{10}|b|) + 80$ terms at dps 200 to 300. Negative bases converge as alternating series with no modification.

The relation lattice R . Complex conjugation splits the problem exactly. With $\rho_s = \text{Re } L_s$ symmetric under $s \rightarrow N - s$ and $\sigma_s = \text{Im } L_s$ antisymmetric, any c decomposes as $c_+ + c_-$ with $\sum c_s \rho_s = \sum c_+ \rho_s$ and $\sum c_s \sigma_s = \sum c_- \sigma_s$. So R is the direct sum of a symmetric and an antisymmetric part, each found by PSLQ on about $N/2$ reals rather than a $2N+2$ dimensional lattice reduction. LLL at that size did not terminate.

Complete relation sets. Recursive elimination: find one relation by PSLQ, drop the coordinate where the relation is largest, recurse. This is complete, because any further relation reduces modulo the found one to a relation supported off that coordinate. Residual gate 10^{-230} at dps 300; results stable from maxcoeff 10^5 to 10^7 .

Kernel realisation over a named field. Given w in R , the realising polynomial is unique up to scale by inverse DFT: $c_r = \lambda w_{\text{hat}_r} / (N \theta^r)$ with $w_{\text{hat}_r} = \sum_j w_j \omega^{(-jr)}$. Solvability of $C(o) = 0$ is exactly $\sum_j w_j = 0$ by Lemma 3.1. Membership of c_r in a named field is tested by PSLQ against that field's power basis. The phase λ must be free: 1 for the symmetric part and i for the antisymmetric part.

Exact rational kernel dimension. To avoid basis-dependent counting, the kernel is obtained by solving the rational system directly in $\mathbb{Z}[\zeta_N]$: for each u in the orthogonal complement of R , $\sum_r c_r \theta^r T_r(u) = 0$ expands into $\deg(Q(\zeta_N))$ rational equations, and the nullspace is computed symbolically. At (4096,24) this is 136 equations in 23 unknowns and returns 5 exactly.

Appendix B. Code

B.1 Deriving a seed kernel without search. The forcing lemma supplies the vanishing set; the remaining conditions are linear.

```
import sympy as sp
y, a, b, c, lam = sp.symbols('y a b c lam')

# forced: the two unshared-prime orbits, plus y | C
C8 = sp.expand(y*(y**2+2)*(y**2+2*y+2)*(a + b*y + c*y**2))
root = {0: sp.sqrt(2), 4: -sp.sqrt(2), 1: 1+sp.I, 7: 1-sp.I}

eqs = [sp.Eq(sp.simplify(C8.subs(y, root[0])), lam),
            sp.Eq(sp.simplify(C8.subs(y, root[4])), lam),
            sp.Eq(sp.simplify(C8.subs(y, root[1])), -lam),
            sp.Eq(sp.simplify(C8.subs(y, root[7])), -lam)]
sol = sp.solve(eqs, [a, b, c, lam], dict=True)[0]
# one free parameter remains -> dim K = 1 forced
# coefficients: [8, -8, -4, -8, -2, -2, 1]
```

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B.2 The exclusion sieve. Factorisation only.

```
from sympy import factorint
import sympy as sp
x = sp.Symbol('x')

def sieve(b, n):
    orbits = [sp.Poly(g, x) for g, _ in sp.factor_list(x**n - b)[1]]
    simple = {p for p, e in factorint(b-1).items()
              if e == 1 and p != 2 and b % p != 0 and n % p != 0}
    forced = set()
    for i, gp in enumerate(orbits):
        g1 = int(gp.eval(1))
        if abs(g1) <= 1: continue
        for p in factorint(abs(g1)):
            if p in simple: forced.add(i)
    deg = sum(orbits[i].degree() for i in forced)
    return 1 + deg <= n - 1    # necessary for dim K > 0
```

B.3 The relation lattice by conjugation split.

```
import mpmath as mpm

def relation_lattice(b, N, dps=280):
    mpm.mp.dps = dps
    th = mpm.mpf(abs(b))**(mpm.mpf(1)/N)
    ph = 0 if b > 0 else mpm.pi/N    # negative bases: half-step offset
    al = [th*mpm.exp(1j*(2*mpm.pi*s/N + ph)) for s in range(N)]
    L = [mpm.log(1 - 1/a) for a in al]
    h = N//2
    w = [1 if (s == 0 or 2*s == N) else 2 for s in range(h+1)]
    sym = all_relations([w[s]*mpm.re(L[s]) for s in range(h+1)])
    anti = all_relations([2*mpm.im(L[s]) for s in range(1, h)])
    return sym, anti    # dim R = len(sym) + len(anti)

def all_relations(v, maxcoeff=10**6, tol=None):
    n = len(v)
    if n < 2: return []
    if tol is None: tol = mpm.mpf(10)**(-int(mpm.mp.dps*0.7))
    r = mpm.pslq(v, maxcoeff=maxcoeff, maxsteps=25000, tol=tol)
    if r is None: return []
    if abs(sum(int(a)*t for a, t in zip(r, v))) > tol*mpm.mpf(10)**10:
        return []
    i = max(range(n), key=lambda k: abs(r[k]))
    out = [list(map(int, r))]
    for s in all_relations([v[k] for k in range(n) if k != i], maxcoeff, tol):
        s = list(s); s.insert(i, 0); out.append(s)
    return out
```

B.4 The gate. Occupancy read as bins modulo the seed period.

```
def occupied_bins(b, N, seed_n):
    sym, anti = relation_lattice(b, N)
```

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```

R = []
h = N//2
for a in sym:
    c = [0]*N
    for s in range(h+1):
        c[s] += a[s]
        if 0 < s < N-s: c[N-s] += a[s]
    R.append(c)
for a in anti:
    c = [0]*N
    for i, s in enumerate(range(1, h)):
        c[s] += a[i]; c[N-s] -= a[i]
    R.append(c)
occ = set()
for c in R:
    occ |= {j % seed_n for j in range(N) if c[j] != 0}
return len(R), sorted(occ)

# (16,24) -> {0,1,2,4,6,7} 3-channel open
# (16,40) -> {0,1,3,4,5,7} 5-channel open
# (16,56) -> {0,1,4,7} nothing open

```

B.5 The inverse procedure, with the equimodularity test.

```

import sympy as sp, mpmath as mpm
from sympy import factorint
y = sp.Symbol('y')

def invert(c, N, tol=mpm.mpf(10)**-20):
    C = sum(c[r-1]*y**r for r in range(1, len(c)+1))
    orbits, free = [], []
    for g, _ in sp.factor_list(C[1]):
        roots = sp.Poly(g, y).all_roots()
        mods = [mpm.mpf(abs(complex(sp.N(r, 30)))) for r in roots]
        if all(m < tol for m in mods):
            continue # the factor y: r = 0 exclusion
        g1 = int(sp.expand(g).subs(y, 1))
        ps = sorted(p for p in factorint(abs(g1)) if p > 2 if abs(g1) > 1 else [])
        equimodular = max(mods) - min(mods) < tol
        on_unit = abs(mods[0] - 1) < tol
        if equimodular and not on_unit:
            orbits.append((g, mods[0], ps))
        else:
            free.append((g, ps)) # discarded even if it carries primes
    rho = orbits[0][1]
    U = 1
    for _, _, ps in orbits:
        for p in ps: U *= p
    return {'rho': rho, 'abs_b': rho**N, 'U': U,
            'free_primes': [p for _, ps in free for p in ps]}

invert([8,-8,-4,-8,-2,-2,1], 8) # rho 1.4142136, |b| 16, U 15
invert([16,-24,-8,-6,1], 6) # rho 2.0, |b| 64, U 7, free carries [3]
# note: U is the UNshared part of b-1. at hex, b-1 = 63 and U = 7.

```

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Appendix C. Measured Frames

dim R measured by conjugation-split iterated PSLQ; incidence m_p by factorisation. An asterisk marks p dividing N.

frame	orbits	dim R	incidence m_p
(4,6)	2	1	3:1*
(4,12)	2	1	3:1*
(16,8)	4	1	3:1 5:1
(16,12)	3	1	3:1* 5:1
(16,16)	4	1	3:1 5:1
(16,24)	4	5	3:1* 5:1
(16,32)	4	1	3:1 5:1
(16,40)	4	5	3:1 5:1*
(16,56)	4	1	3:1 5:1
(64,6)	4	1	3:2 7:1
(64,12)	4	2	3:2* 7:1
(-64,12)	4	3	2:- 5:1 13:1
(64,18)	4	3	3:2* 7:1
(64,24)	4	4	3:2* 7:1
(64,42)	4	1	3:2* 7:1*
(81,12)	3	0	5:1
(256,16)	5	1	3:1 5:1 17:1
(4096,12)	6	1	3:2* 5:1 7:1 13:1
(4096,24)	8	7	3:2* 5:1 7:1 13:1

Appendix D. Scope

dim R is called frame-intrinsic on the strength of Baker's theorem, which fixes it against widening on the logarithm side. That is a theorem. What is not established is that R as computed here — the Q-relation lattice

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among principal-branch logarithms — exhausts every analytic relation the frame supports. It exhausts the Qbar-linear ones. No claim is made beyond that.

The sieve's completeness is bounded by its stated domain: b in $[2, 320]$, n in $[3, 9]$, positive bases. The negative half is unswept and has already produced a seed. No global classification is claimed anywhere in this paper.

The gate is verified on five octal frames and has a counterexample on the hexagonal lineage at $(64, 42)$. It is stated as an octal-lineage law. The lineage character result rests on six octal and five hexagonal measurements; it is a measured contrast, not a theorem.

Framework language used during this work — relation as primitive, absence as operator, reader versus object — is interpretation. It appears only in section 1 as motivation for the measurements and is not load-bearing for any result here.

References

Bailey, D. H., Borwein, P. B., Plouffe, S. (1997). On the rapid computation of various polylogarithmic constants. *Mathematics of Computation* 66(218), 903–913.

Borwein, J. M., Borwein, D., Galway, W. F. (2004). Finding and excluding b -ary Machin-type individual digit formulae. *Canadian Journal of Mathematics* 56, 897–925.

Baker, A. (1966). Linear forms in the logarithms of algebraic numbers. *Mathematika* 13, 204–216.

Ferguson, H. R. P., Bailey, D. H., Arno, S. (1999). Analysis of PSLQ, an integer relation finding algorithm. *Mathematics of Computation* 68(225), 351–369.

Washington, L. C. (1997). *Introduction to Cyclotomic Fields*, 2nd ed. Springer.

Kulik, D. (2026). Kernel, Frame, Winding, Residue. QuHarmonics Research Group.

Kulik, D. (2026). The Reader and the Frame. QuHarmonics Research Group.